

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	17
Experiment Name	Detect Outliers in a Given Dataset

DESCRIPTION

To perform the experiment: Detect Outliers in a Given Dataset.

CODE

```
import numpy as np
import matplotlib.pyplot as plt

data = np.array([10, 12, 13, 14, 15, 16, 18, 19, 20, 100])
q1, q3 = np.percentile(data, [25, 75])
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
outliers = data[(data < lower) | (data > upper)]
print("Q1:", q1, "Q3:", q3, "IQR:", iqr)
print("Lower limit:", lower)
print("Upper limit:", upper)
print("Outliers:", outliers)
plt.boxplot(data)
plt.title("Outlier Detection Using Box Plot")
plt.ylabel("Value"); plt.show()
```

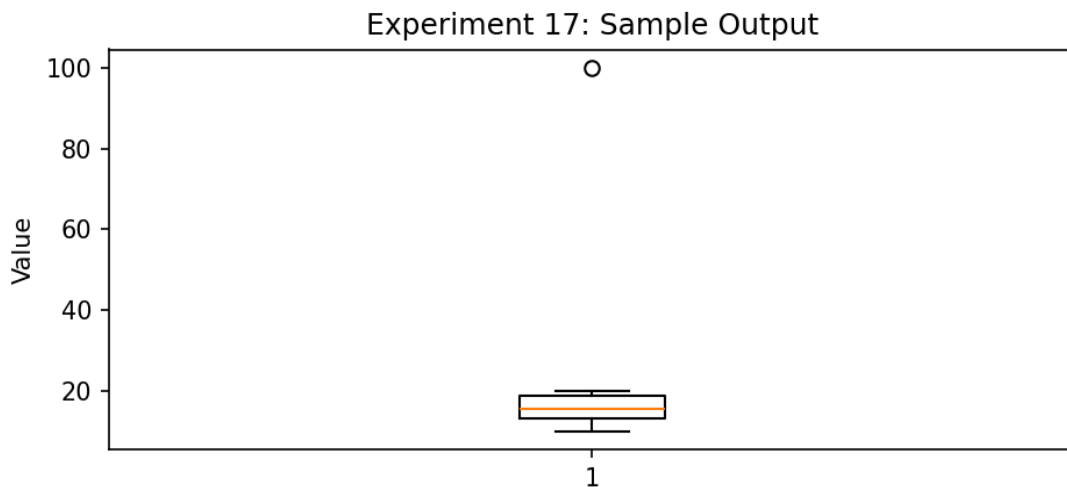
SAMPLE INPUT

Sample input: [10, 12, 13, 14, 15, 16, 18, 19, 20, 100].

OUTPUT — SAMPLE CONSOLE OUTPUT

IQR rule applied.\nOutlier detected: 100

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature